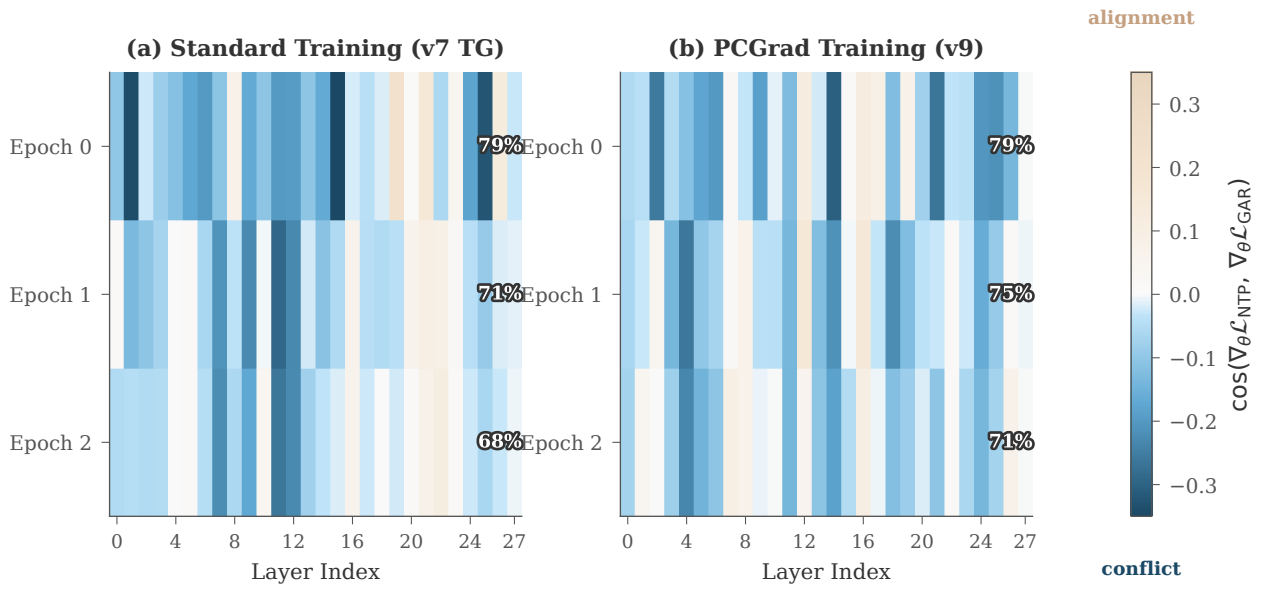
(a) Standard Training (v7 TG)
(b) PCGrad Training (v9)
Epoch 0
Epoch 1
Epoch 2
79%
71%
68%
79%
75%
71%
0
4
8
12
16
20
24
27
Layer Index
alignment
conflict
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
$\cos(\nabla_\theta \mathcal{L}_{\text{NTP}}, \nabla_\theta \mathcal{L}_{\text{GAR}})$